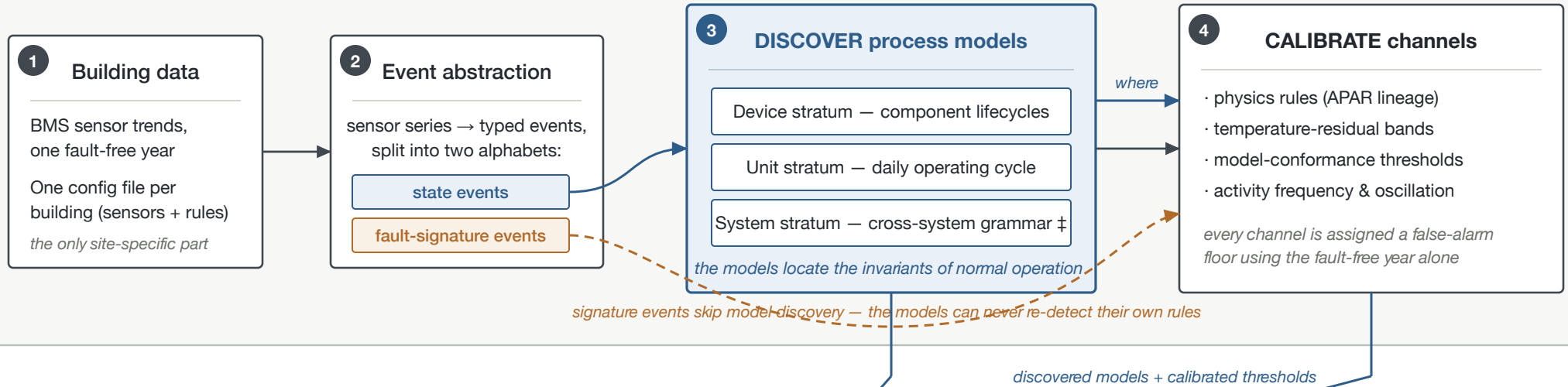
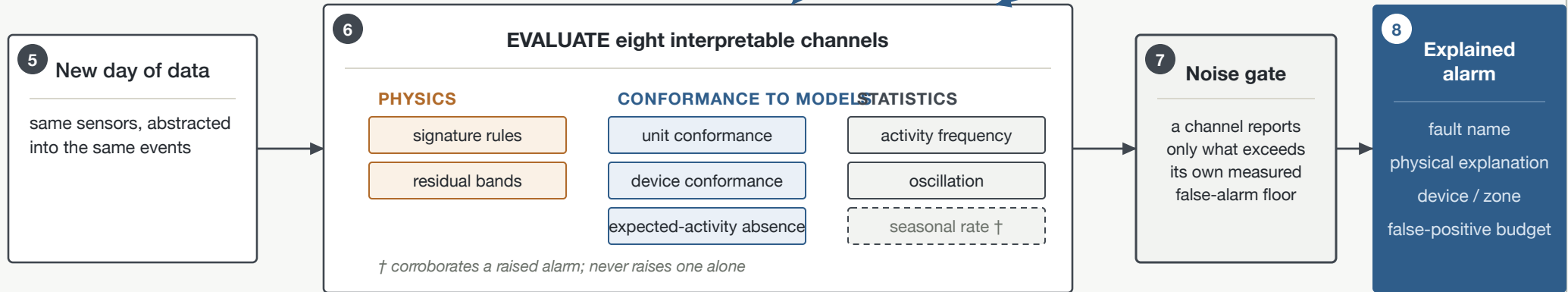


A TRAINING — learn normal behaviour from one fault-free year



B DEPLOYMENT — score each new day



Validated on three LBNL benchmark systems — 13/14 · 23/30 · 24/29 fault scenarios detected, median time-to-detect 1 day, joint false-alarm budget 1.0–5.2%.

□ discovered process models □ physics / fault-signature path - - - excluded from discovery
‡ supports the portability analysis; the deployed channels come from the device and unit strata